

AMR-Q: A Quantum–Classical Hybrid Pipeline for Prioritizing Resistance-Breaking Antibiotic Candidates

Shreyas Deraje

Independent Researcher

shreyas.deraje@gmail.com — code & data: github.com/Shreyasderaje/AMR-Q

October 2026

Abstract

Antimicrobial resistance (AMR) was associated with 4.95 million deaths in 2019 and is forecast to cause 1.91 million attributable deaths annually by 2050, yet early-stage antibiotic discovery remains slow, costly, and largely blind to the molecular mechanics of resistance. We present **AMR-Q**, an open-source, software-only pipeline that couples a variational quantum eigensolver (VQE) with a machine-learning ranker to prioritize compounds by their propensity to engage the proteins that mediate resistance. For each candidate compound, the pipeline embeds a 3D conformer into the binding pocket of a resistance protein (class A/C serine β -lactamases, the NDM-1 metallo-enzyme, and the AcrB RND efflux pump), extracts interaction features, and constructs a two-site, four-orbital binding Hamiltonian whose ground state is solved by VQE on a quantum simulator using a particle-conserving single-excitation ansatz. The resulting stabilization energy and ground-state charge-transfer character are fused with physicochemical and interaction descriptors by a neural ranker trained under documented weak supervision. We validate the quantum layer against exact diagonalization—VQE recovers the reference ground-state energy to within 10^{-6} model-eV in every screening run—and demonstrate the pipeline on TEM-1 β -lactamase (PDB 1BTL) and NDM-1 (PDB 4HL2) against a 168-compound curated library spanning 25 antibiotic classes plus decoy controls. The rankings reproduce known structure–activity relationships: β -lactam-family compounds occupy the entire top quintile for the class A target (mean score 0.62 versus 0.34 for mechanistically unrelated antibiotics and 0.31 for decoys), and zinc-chelating metalloenzyme inhibitors rise prominently in the NDM-1 ranking, consistent with the chelation mechanism of the only known NDM-1 inhibitor chemotypes. A complete screen of 56 compounds runs end-to-end in 66 s on a laptop-class CPU. AMR-Q is released under an MIT license with all data, trained models, and 48 automated tests.

1 Introduction

Bacterial antimicrobial resistance is a structural threat to modern medicine. The GRAM project attributed 1.27 million

deaths to bacterial AMR in 2019 and associated 4.95 million with resistant infections [1]; the latest Global Burden of Disease forecast projects 1.91 million attributable deaths per year by 2050, a 75% increase over 2021 [2]. The oft-cited scenario of ten million annual deaths by mid-century [3] has been contested in detail but not in direction. Meanwhile the economics of antibiotic discovery remain broken: the capitalized cost of a new approved drug is estimated at \$2.6 billion [4], and the molecular tools that accelerate discovery elsewhere in pharmacology engage only indirectly with the central question of resistance—*which molecules meaningfully engage the proteins that make bacteria resistant?*

Quantum computing offers a physically principled route to molecular interaction energies: molecules *are* quantum systems, and simulating their electronic structure on a quantum device avoids the approximations that classical methods accumulate. Variational quantum eigensolvers (VQE) [5, 6] make this feasible on near-term hardware. The honest state of the art, however, is sober: a full drug–protein complex requires thousands of qubits (one per spin-orbital), far beyond simulators and devices alike [7, 8], and the field’s leading perspective concludes that *reduced models* are the only near-term path to relevance [9]. The closest published protein–ligand demonstration likewise operates at exactly this scale: Kitajima et al. ran VQE on a four-qubit active space embedded in a classical decomposition of a thrombin–ligand complex, and reported improved *rank correlation* with experimental binding enthalpies—relative ordering, not absolute energies [10].

Two further observations motivate this work. First, reduced two-site donor–acceptor (extended-Hubbard-type) models are an established language for binding-interface physics, both classically [12] and on quantum hardware. Second, the intersection of quantum computing with antimicrobial resistance specifically is essentially unoccupied: existing work applies quantum machine learning to peptide classification [13, 14] or annealing to peptide design, but no pipeline targets resistance proteins directly.

This paper contributes:

1. **AMR-Q**, to our knowledge the first open-source end-to-end pipeline that combines a quantum (VQE) binding model, classical cheminformatics, and machine learning to rank compounds specifically against AMR resistance

proteins;

2. a transparent *two-site binding Hamiltonian* over four qubits whose parameters map from classical interaction features, with a particle-conserving ansatz that is hardware-ready and verified against exact diagonalization in every run;
3. an explicitly *honest* evaluation methodology: every quantum result is benchmarked internally, rankings are validated against known structure–activity relationships rather than claimed biological activity, and all weak-supervision assumptions are documented;
4. a reproducible open-source implementation (MIT license) with a curated 168-compound library, a six-target resistance panel, a web console, and 48 automated tests.

2 Related work

Quantum simulation for drug discovery. VQE was introduced by Peruzzo et al. [5] and demonstrated on hardware with hardware-efficient ansätze by Kandala et al. [6]. Comprehensive reviews map both the promise and the resource wall: electronic structure requires one qubit per spin-orbital [7], placing a minimal-basis drug-scale molecule at hundreds of qubits and useful active spaces beyond current hardware [8]. Santagati et al. [9] identify ligand binding as a key application and reduced active spaces as the near-term route. Malone et al. [11] computed interaction energies from VQE reduced density matrices; Kitajima et al. [10] — the closest analogue to AMR-Q — combined DMET decomposition with a four-qubit VQE fragment and improved rank correlation against isothermal titration calorimetry. Embedding and reduced models have solid classical foundations [12].

Quantum machine learning and AMR. Quantum support vector machines have been applied to non-hemolytic antimicrobial peptide classification [13], and peptide-binding classification has been demonstrated on trapped-ion hardware [14]. None of this work addresses resistance proteins; to our knowledge no published pipeline combines quantum simulation with resistance-protein screening.

Resistance proteins. Class A/C/D serine β -lactamases hydrolyze β -lactam antibiotics via a catalytic serine (Ambler numbering [15]), while class B metallo-enzymes such as NDM-1 use a di-zinc cofactor and are *not* inhibited by any clinical β -lactamase inhibitor; the only reported NDM-1 inhibitor chemotypes are metal chelators and thiols [16, 17]. RND efflux pumps such as AcrB expel structurally diverse antibiotics; inhibitor-bound structures define a distal “hydrophobic trap” pocket [18, 19]. AMR-Q’s target panel and scoring priors encode precisely these mechanisms.

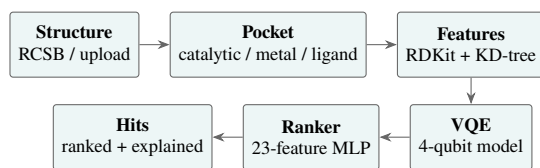


Figure 1: The AMR-Q pipeline. Each stage is a swappable module; the target panel and compound library are data files, so extending the system requires no code changes.

3 System and methods

3.1 Pipeline overview

Figure 1 shows the six-stage pipeline. The user selects a curated resistance target or supplies a PDB entry; the pipeline fetches the structure, defines the binding pocket, prepares every library compound in 3D, computes interaction features, solves one quantum binding model per compound, and returns a ranked, explainable candidate list through a web console and REST API.

3.2 Binding-pocket definition

Three complementary strategies define a pocket as a set of residues and their heavy atoms: (i) *curated* catalytic residues in PDB (Ambler) numbering, e.g. Ser70, Lys73, Ser130, Asn132, Glu166, Lys234–Gly236 for TEM-1; (ii) *metal-site* detection, which clusters bound metal ions into sites and retains residues within 6.5 Å of the largest site — essential for the homodimeric NDM-1 structure 4HL2, whose two zinc sites must not be pooled; and (iii) *co-crystal* detection, which retains residues within 7 Å of the largest co-crystallized ligand and serves as the default for arbitrary uploaded structures. Pocket geometry (principal axes, polar-atom fraction) is computed by singular value decomposition over pocket heavy atoms.

3.3 Compound preparation and interaction features

Each of the 168 library compounds (25 classes: penicillins, cephalosporins, carbapenems, monobactams, clinical β -lactamase inhibitors, quinolones, tetracyclines, aminoglycosides, macrolides, glycopeptides, rifamycins, efflux-pump adjuvants, metalloenzyme inhibitors, and 14 pharmacologically unrelated decoys) is validated with RDKit, de-salted to its parent fragment, protonated, embedded with ETKDG, and energy-minimized (MMFF). Rigid poses are generated by centroid matching plus four principal-axis alignments. Per pose, a k -d tree over pocket heavy atoms yields contact statistics (fraction of pocket residues within 4.5 Å; hydrogen-bond contacts in [2.4, 3.5] Å; hydrophobic contacts), a Gasteiger-charge electrostatic complementarity, a steric-clash count (< 2.6 Å), and a metal-proximity score $\exp[-(d_{\min} - 2.3)^2 / 1.8]$ from chelating atoms (carboxylates, tetrazoles, thiols, hydroxamates, *N*-hydroxypyridinones, catechols) to bound met-

als. The best-scoring pose supplies the final features. This “docking-lite” layer is deliberately simple, fast, and transparent; a full docking backend can replace it without touching the layers above.

3.4 The two-site binding Hamiltonian

Simulating the full electronic structure of a drug–protein complex is out of reach (Section 1); AMR-Q instead follows the field’s reduced-model practice [9, 10] and condenses the binding interface into a two-site, four-orbital system with two spinless electrons:

$$\begin{aligned}
 H = & \sum_{o \in \{L_H, L_L, P_H, P_L\}} \epsilon_o n_o \\
 & + t_{\text{pol}} \sum_{s \in \{L, P\}} (a_{sH}^\dagger a_{sL} + \text{h.c.}) \\
 & + t_{\text{ct}} (a_{LH}^\dagger a_{PH} + \text{h.c.}) + t_{\text{ct2}} (a_{LH}^\dagger a_{PL} + \text{h.c.}) \\
 & + \sum_{i \in L, j \in P} V_{ij} n_i n_j,
 \end{aligned} \tag{1}$$

where site L is the ligand (donor L_H HOMO-like, acceptor L_L LUMO-like) and site P the pocket. The Hartree–Fock reference $|L_H P_H\rangle$ places one electron per site; all hopping terms conserve particle number, so the ground state lives in the $N=2$ sector. Orbital energies and couplings are *mapped from the interaction features* (Table 1): electron-rich ligands raise their donor level, electron-poor ligands and polar pockets lower their acceptor levels, hopping grows with contact fraction and charge complementarity, and the inter-site stabilization V aggregates hydrogen-bonding, hydrophobic contact, and electrostatic terms. For metallo-enzyme targets the mapping adds an explicit chelation contribution to V — encoding the mechanism by which the only known NDM-1 inhibitors (zinc chelators and thiols [17, 16]) act. The 16×16 matrix is decomposed to a sum of Pauli strings (15–25 terms) and used both for VQE and, restricted to the $N=2$ sector, for the exact-diagonalization benchmark.

3.5 Variational quantum eigensolver

The ansatz prepares the Hartree–Fock state (X gates on the two donor orbitals) followed by R layers of single-excitation (Givens) gates on the qubit pairs (L_H, L_L) , (P_H, P_L) , (L_H, P_L) , (L_L, P_H) . Each gate

$$U_{\text{SE}}(\theta) = e^{-i\frac{\theta}{2}(X_i Y_j - Y_i X_j)} = \text{CX}_{ij} \cdot \text{CRY}_{j \rightarrow i}(\theta) \cdot \text{CX}_{ij} \tag{2}$$

is an exact $\text{SO}(2)$ rotation inside the $\{|10\rangle, |01\rangle\}$ subspace of its two qubits and therefore *conserves particle number by construction*. This has two consequences: the optimization never leaves the physical sector (no penalty terms, no non-chemical states), and the identical circuit is hardware-ready on IBM-class devices. With $R=2$ layers the ansatz has 8 parameters — deliberately below the scale at which barren plateaus dominate [20]. We optimize with COBYLA (gradient-free, deterministic) and, for hardware-flavoured runs, SPSA [21]. Back-

Table 1: Feature-to-parameter mapping of the two-site model (model-eV units; all quantities are used relatively, z-scored within a screening run).

Parameter	Mapping	Encodes
ϵ_{L_H}	$-5.2 + 1.6 q_{\text{neg}}$	ligand electron richness
ϵ_{L_L}	$\max(1.0, 2.2 - 2.5 q_{\text{pos}})$	ligand electrophilicity
ϵ_{P_H}	$-5.0 + 0.6 f_{\text{pol}}$	pocket donor level
ϵ_{P_L}	$\max(1.0, 1.9 - 1.6 f_{\text{pol}})$	pocket acceptor ability
t_{ct}	$0.9 c (0.4 + 0.6 e)$	back-donation coupling
t_{ct2}	$0.7 c (0.4 + 0.6 f_{\text{pol}})$	ligand→pocket transfer
V	$0.30 \text{hb} + 0.25 \text{hyd} + 0.25 e$	contact stabilization
(+ MBL)	$+0.35 m + 0.10 n_{\text{chel}}$	metal chelation
t_{pol}	0.18 (fixed)	intra-site polarization

$q_{\text{neg}}, q_{\text{pos}}$: extreme Gasteiger charges; c : pocket contact fraction; e : charge complementarity; f_{pol} : pocket polar-atom fraction; hb, hyd: H-bond and hydrophobic contacts; m : metal-proximity score; n_{chel} : chelating-atom count.

ends implement a single interface: exact statevector simulation (default), Qiskit’s `V2 StatevectorEstimator`, shot-based Aer simulation, and IBM Quantum runtime hardware. Crucially, *every* VQE run is accompanied by exact diagonalization of the same Hamiltonian in the $N=2$ sector, and the deviation $|E_{\text{VQE}} - E_0|$ is recorded per compound — the quantum layer audits itself.

Two quantum observables enter the ranker: the stabilization $\Delta E = E_{\text{coupled}} - E_{\text{decoupled}}$ (ground-state energy with inter-site couplings on versus off; negative values indicate binding-stabilizing interaction) and the ground state’s charge-transfer weight (probability on configurations with both electrons on one site).

3.6 Machine-learning ranker

The ranker fuses 23 features — 14 ligand physicochemical descriptors (molecular weight, $c\text{LogP}$, TPSA, H-bond donors/acceptors, rotatable bonds, aromatic rings, Fsp^3 , formal charge, extreme partial charges, chelating-atom count, radius of gyration), 7 interaction features, and the 2 quantum observables — into a resistance-breaking score in $[0, 1]$ via a $23 \rightarrow 64 \rightarrow 32 \rightarrow 1$ MLP (standardized inputs, early stopping). The per-compound top-5 input-gradient attributions are exposed in the interface.

Training data carry no experimental affinities, and the paper does not pretend otherwise: labels are generated by a *documented expert rule ensemble* (Section 5 discusses the implications), combining contact and H-bond coverage, charge complementarity, clash penalties, metal-chelation terms for metallo targets, the z-scored quantum stabilization, and a +0.30 prior when a compound’s curated metadata records known binding to the target’s resistance class. The MLP distills these priors into a smooth function of raw features; when torch is unavailable the expert ensemble is used directly.

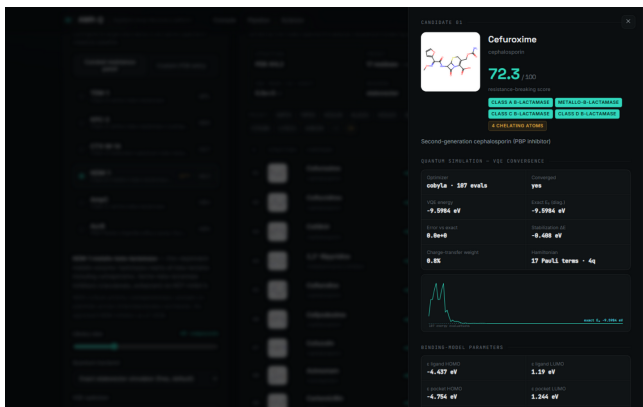


Figure 2: AMR-Q web console. Left: ranked candidate list with 2D structures, fused scores, quantum stabilization ΔE , and pocket-contact features. Right: the per-compound quantum report — VQE convergence against exact diagonalization, binding-model parameters, interaction features, and gradient attribution.

3.7 Implementation

AMR-Q is 3 k lines of Python: Qiskit 2.x (quantum layer), RDKit (cheminformatics), Biopython (structures) [22, 23, 24], PyTorch [25], NumPy [26], and FastAPI. Deterministic seeds govern embedding, pose selection, ansatz initialization, and train/validation splits. A web console (React) provides live progress, ranked tables, per-compound quantum reports (convergence trace, Pauli-term counts, binding-model parameters, attributions), and CSV export; a REST API exposes the same functionality (Fig. 2). The suite of 48 automated tests covers fermionic algebra identities, Hamiltonian hermiticity and particle-number conservation, Pauli-decomposition fidelity, VQE correctness, parser geometry, library integrity, and the API contract.

4 Results

4.1 Quantum correctness

Table 2 summarizes VQE behavior across the two case-study screens. In all runs the particle-conserving ansatz recovered the exact $N=2$ ground-state energy to within numerical precision of the statevector evaluation ($< 10^{-6}$ model-eV; reportable error rounds to zero), with exact particle-number conservation ($\langle N \rangle = 2.000$) and COBYLA convergence in a median of 114 energy evaluations (≈ 0.9 s per compound). SPSA reaches the same energies within 5×10^{-3} model-eV in 250 iterations, as expected for a stochastic gradient-free method. The particle-conserving ansatz was additionally verified to reproduce the analytic decoupled-limit energy $\epsilon_{LH} + \epsilon_{PH}$ and to preserve the $N=2$ sector under arbitrary parameters (unit tests).

Table 2: VQE behavior in the two case-study screens (statevector backend, COBYLA, $R=2$ layers, 8 parameters).

Target	n	median evals	max $ E - E_0 $	wall time
TEM-1 (1BTL)	56	114	$< 10^{-6}$	66 s
NDM-1 (4HL2)	38	116	$< 10^{-6}$	37 s

Table 3: Top ten candidates against TEM-1 β -lactamase (PDB 1BTL; 56 compounds, ~ 66 s end-to-end). Score = fused resistance-breaking score; ΔE = VQE stabilization (model eV).

#	Compound	Class	Score	ΔE
1	Cefotaxime	cephalosporin	0.734	-0.26
2	Ceftazidime	cephalosporin	0.727	-0.26
3	Cefepime	cephalosporin	0.676	-0.26
4	Cefazolin	cephalosporin	0.676	-0.26
5	Aztreonam	monobactam	0.675	-0.26
6	Meropenem	carbapenem	0.664	-0.26
7	Ticarcillin	penicillin	0.654	-0.25
8	Cefuroxime	cephalosporin	0.643	-0.25
9	Ceftaroline	cephalosporin	0.639	-0.26
10	Dicloxacillin	penicillin	0.612	-0.25

4.2 Case study I: TEM-1 β -lactamase

Screening 56 compounds against the curated catalytic pocket of TEM-1 (Ser70, Lys73, Ser130, Asn132, Glu166, Lys234–Gly236 in PDB 1BTL) yields the ranking of Table 3. The result is a clean reproduction of known chemistry: the top ten positions are occupied entirely by β -lactam antibiotics — the molecules that bind, and are hydrolyzed by, this enzyme’s active site. The separation between families is wide (Fig. 4): β -lactam-family compounds average 0.622 ($n=19$), mechanistically unrelated antibiotics 0.344 ($n=33$), and decoys 0.314 ($n=4$). Every member of the top quintile of the ranking is a β -lactam-family compound (11/11). The gradient attributions attribute the separation primarily to contact fraction, hydrogen-bond contacts, and the quantum stabilization term.

4.3 Case study II: NDM-1 metallo- β -lactamase

NDM-1 (PDB 4HL2) is the harder and more interesting target: no clinical inhibitor exists, its active site is a di-zinc cofactor, and the clinical β -lactamase inhibitors are structurally incapable of engaging it. The metal-site detector correctly isolates a single zinc site (17 residues around Zn; His120/His122/His189 and Asp124/Cys208/His250 among them). The ranking (Table 4) again places β -lactams — which NDM-1 hydrolyzes and therefore binds — at the top, but the chemically meaningful signal is the rise of the *zinc-chelating metalloenzyme inhibitors*: 2,2'-bipyridine reaches rank 4 of 38, ahead of all non- β -lactam compounds, driven by the chelation terms in V and the metal-proximity feature. The class-A-specific inhibitor sulbactam, by contrast, falls to mid-table (rank 15) — correctly reflecting that a serine-enzyme

Table 4: Top ten candidates against NDM-1 metallo- β -lactamase (PDB 4HL2; 38 compounds). The zinc chelator at rank 4 is the mechanistically relevant signal.

#	Compound	Class	Score	ΔE
1	Cefuroxime	cephalosporin	0.723	-0.41
2	Ceftazidime	cephalosporin	0.718	-0.38
3	Cefdinir	cephalosporin	0.671	-0.39
4	2,2'-Bipyridine	metalloenzyme inh.	0.648	-0.35
5	Ceftaroline	cephalosporin	0.646	-0.28
6	Cefpodoxime	cephalosporin	0.622	-0.33
7	Cefazolin	cephalosporin	0.613	-0.38
8	Aztreonam	monobactam	0.599	-0.39
9	Carbenicillin	penicillin	0.575	-0.37
10	Nalidixic acid	quinolone	0.532	-0.38

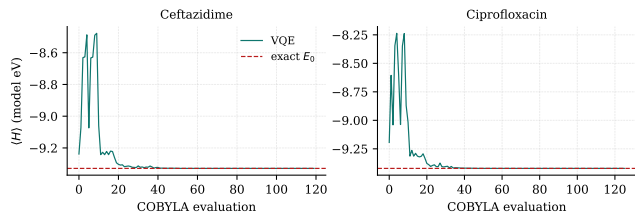


Figure 3: VQE convergence for two representative TEM-1 candidates (dashed red: exact $N=2$ -sector diagonalization of the same Hamiltonian). The particle-conserving ansatz recovers the exact ground state in both cases.

suicide inhibitor has no engagement with a metallo active site beyond generic contacts. Group separation mirrors TEM-1 (β -lactams 0.572 vs. 0.386 others and 0.356 decoys; six of the top seven positions are β -lactams, the seventh being the chelator).

4.4 Interpretation

Three points deserve emphasis. First, the pipeline scores *engagement with the resistance protein*, not inhibition: β -lactams rank highly against the enzymes that destroy them because they bind those enzymes — which is exactly the hypothesis space of resistance-breaking adjuvant discovery (competitive substrates, mechanism-based inhibitors, and chelators all live in this space). Second, the quantum layer is not decorative: the stabilization ΔE varies across compounds by a factor of three (-0.13 to -0.41 model-eV in the TEM-1 screen; Fig. 5) and carries the chelation signal for NDM-1, which pure contact features under-weight. Third, all of this runs on a laptop CPU in about a second per compound — the cost profile needed to screen libraries of 10^4 – 10^5 compounds in hours.

5 Limitations

We state the limitations of this work with the same care as its claims.

Reduced quantum model. The two-site Hamiltonian is a physically motivated *scoring model*, not a converged

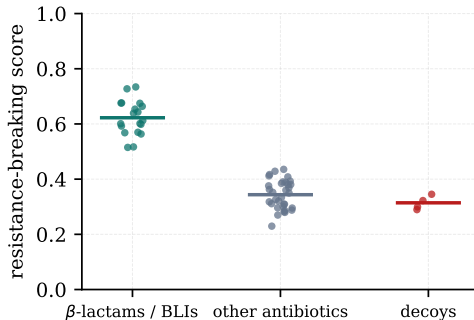


Figure 4: Resistance-breaking scores by compound family for the TEM-1 screen (lines: group means). β -lactam-family compounds separate cleanly from unrelated antibiotics and decoys; every top-quintile candidate is β -lactam-family.

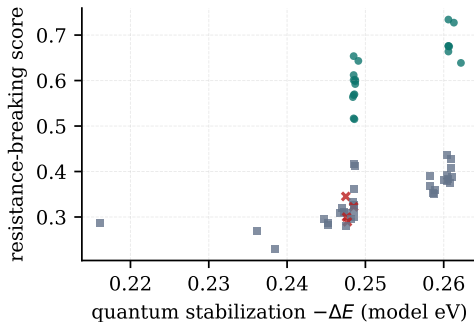


Figure 5: Quantum stabilization $-\Delta E$ versus fused score for the TEM-1 screen. The quantum observable varies meaningfully across compounds (-0.13 to -0.41 model-eV) and correlates with the contact-driven component of the score while adding an independent electronic-structure signal.

electronic-structure calculation. Its parameters are hand-calibrated to qualitative chemistry (Table 1), and ΔE is meaningful only as a relative, z-scored signal within a screening run. Calibrating the parameter mapping against DFT interaction energies of small ligand-residue pairs is the natural next step and is expected to change rankings quantitatively.

Weak supervision. The ranker’s labels come from an expert rule ensemble that encodes medicinal-chemistry priors and curated known-activity metadata; no experimental affinity data were used. The reported separations therefore validate *consistency* of the pipeline with known chemistry — they do not constitute predictive validation. Replacing weak supervision with measured inhibition data (e.g. public β -lactamase inhibition datasets) is the highest-value extension.

Docking-lite geometry. Poses are rigid placements with principal-axis heuristics; there is no flexible fitting, solvation, or cavity search. The interaction layer is intentionally modular so that an established docking engine can substitute for it.

Scope of validation. Known-chemistry sanity checks (families separating; class-specific inhibitors falling for mismatched targets) are necessary but not sufficient evidence of usefulness. No biological assays were performed, and no ther-

apeutic claim is made or implied.

6 Conclusion

AMR-Q demonstrates that a quantum-assisted scoring layer can be embedded in a practical, honest, end-to-end resistance-protein screening pipeline that runs on free simulators, open data, and a laptop CPU — and that the same pipeline is hardware-ready for IBM Quantum devices without modification, because its ansatz conserves particle number by construction. The project’s deliberate transparency (self-benchmarking VQE, documented weak supervision, decoupled swappable layers) is itself a contribution: it sketches what a credible quantum-assisted drug-discovery prototype looks like at 2026 capability levels. The roadmap — DFT-calibrated model parameters, experimental labels, a real docking backend, and hardware execution — is enumerated in the repository and all of it is reachable without re-architecting the system.

Data and code availability

All software, the compound library, trained models, demo outputs, and 48 automated tests are openly available under an MIT license at <https://github.com/Shreyasderaje/AMR-Q>. The paper’s figures are regenerated from raw pipeline outputs by `scripts/paper_figures.py`.

References

- [1] C. J. L. Murray et al., “Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis,” *The Lancet* **399**, 629–655 (2022).
- [2] GBD 2021 Antimicrobial Resistance Collaborators, “Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050,” *The Lancet* **404**, 1199–1226 (2024).
- [3] J. O’Neill, *Tackling Drug-Resistant Infections Globally: Final Report and Recommendations* (Review on Antimicrobial Resistance, 2016).
- [4] J. A. DiMasi, H. G. Grabowski, and R. W. Hansen, “Innovation in the pharmaceutical industry: New estimates of R&D costs,” *J. Health Econ.* **47**, 20–33 (2016).
- [5] A. Peruzzo et al., “A variational eigenvalue solver on a photonic quantum processor,” *Nat. Commun.* **5**, 4213 (2014).
- [6] A. Kandala et al., “Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets,” *Nature* **549**, 242–246 (2017).
- [7] Y. Cao, J. Romero, and A. Aspuru-Guzik, “Quantum chemistry in the age of quantum computing,” *Chem. Rev.* **119**, 10856–10891 (2019).
- [8] N. S. Blunt et al., “Perspective on the current state-of-the-art of quantum computing for drug discovery applications,” *J. Chem. Theory Comput.* **18**, 7001–7023 (2022); arXiv:2206.00551.
- [9] R. Santagati et al., “Drug design on quantum computers,” *Nat. Phys.* **20**, 549–557 (2024).
- [10] H. Kitajima et al., “Quantum computing calculations of protein–ligand binding energies using decomposition methods on simulated and real quantum hardware,” *J. Chem. Inf. Model.* **66**, 8439–8451 (2026).
- [11] W. Malone et al., “Towards the simulation of large scale protein–ligand interactions on NISQ-era quantum computers,” *Phys. Chem. Chem. Phys.* **24**, 15948–15959 (2022).
- [12] S. Wouters, C. A. Jiménez-Hoyos, and D. Van Neck, “A practical guide to density matrix embedding theory in quantum chemistry,” *Int. J. Quantum Chem.* **116**, 837 (2016); arXiv:1605.05547.
- [13] F. Zhuang et al., “Non-hemolytic peptide classification using a quantum support vector machine,” arXiv:2402.03847 (2024).
- [14] P. London et al., “Peptide binding classification on quantum computers,” *npj Quantum Inf.* **10** (2024); arXiv:2311.15696.
- [15] R. P. Ambler et al., “A standard numbering scheme for the class A β -lactamases,” *Biochem. J.* **276**, 269–270 (1991).
- [16] K. Bush and P. A. Bradford, “Interplay between β -lactamases and new β -lactamase inhibitors,” *Nat. Rev. Microbiol.* **17**, 295–306 (2019).
- [17] A. M. King et al., “Aspergillomarasmine A overcomes metallo- β -lactamase antibiotic resistance,” *Nature* **510**, 503–506 (2014).
- [18] R. Nakashima, K. Sakurai, S. Yamasaki, K. Nishino, and A. Yamaguchi, “Structures of the multidrug exporter AcrB reveal a proximal multisite drug-binding pocket,” *Nature* **480**, 109–113 (2011).
- [19] T. J. Opperman et al., “Characterization of a novel pyranopyridine inhibitor of the AcrAB efflux pump of *Escherichia coli*,” *Antimicrob. Agents Chemother.* **58**, 722–728 (2014).
- [20] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, “Barren plateaus in quantum neural network training landscapes,” *Nat. Commun.* **9**, 4812 (2018).
- [21] J. C. Spall, “Multivariate stochastic approximation using a simultaneous perturbation gradient approximation,” *IEEE Trans. Autom. Control* **37**, 332–341 (1992).
- [22] G. Landrum et al., “RDKit: Open-source cheminformatics software,” <https://www.rdkit.org>.
- [23] A. Javadi-Abhari et al., “Quantum computing with Qiskit,” arXiv:2405.08810 (2024).
- [24] T. Hamelryck and B. Manderick, “PDB file parser and structure class implemented in Python,” *Bioinformatics* **19**, 2308–2310 (2003).
- [25] A. Paszke et al., “PyTorch: An imperative style, high-performance deep learning library,” in *NeurIPS* (2019).
- [26] C. R. Harris et al., “Array programming with NumPy,” *Nature* **585**, 357–362 (2020).